Dissertation / Project / Project Work Title:
AI Powered Chatbot Agent for querying company policies
using RAG and Google Gemini AI

Course No: SEWIZC425T

Course Title: Dissertation

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May 2026

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1. Broad Area of Work

This dissertation delves into Artificial Intelligence and Machine Learning, specifically through the development of the **Maavu Policy Agent**. The project leverages cutting-edge AI technologies to transform how organizations manage and access policy information, making institutional knowledge instantly accessible through intelligent, conversational interfaces.
This project meticulously integrates several advanced domains:

- **Natural Language Processing (NLP) and Conversational AI:** Developing intuitive and intelligent dialogue systems for seamless human-computer interaction.
- **Information Retrieval Systems and Semantic Search:** Architecting sophisticated mechanisms for extracting and understanding relevant data from vast knowledge bases.
- **RAG (Retrieval-Augmented Generation) Architecture:** Implementing a cutting-edge framework that combines retrieval of factual information with generative language models to produce highly informed responses.
- **Full-Stack Web Application Development:** Engineering a robust and scalable web platform to host and deliver the AI-powered functionalities.

2. Background

2.1 The Problem

Modern organizations contend with an overwhelming volume of policy documentation spanning numerous domains like employee conduct, data security, and intellectual property, often spread across disparate formats. This creates a multi-faceted problem where employees struggle with information overload, inefficiently navigating hundreds of pages of dense text, and frequently miss or misunderstand critical details, leading to significant knowledge gaps. Furthermore, the static nature of these traditional documents inherently prevents natural language interaction and deep contextual understanding, thereby escalating the risk of non-compliance across the organization.

2.2 The Solution

The Maavu Policy Agent offers a comprehensive, AI-powered conversational interface designed to address these challenges directly. It empowers employees to articulate their queries using natural language and receive precise, source-cited answers derived directly from official policy documents. The agent intelligently maintains conversational context for seamless follow-up questions, rigorously ensures that all provided information is grounded in authoritative policy, and uniquely supports offline functionality through local embeddings to uphold stringent data privacy standards.

3. Objectives

The objects of this dissertation are as follows:

1. **Democratize Policy Access**
    - Enable all employees to instantly access policy information through natural language queries.
    - Eliminate barriers between employees and critical organizational knowledge.
2. **Improve Information Accuracy**
    - Provide AI-Generated responses grounded in official policy documents.
    - Include source citations with every answer for verification and trust.
3. **Enhanced User Experience**
    - Deliver a modern, intuitive conversational interface.
    - Support contextual follow-up questions and conversational history.
    - Provide Instant Responses with Manual Document Searches
4. **Ensure Data Security & Privacy**
    - Implement secure authentication and authorization.
    - Maintain audit trails of policy queries and access.
5. **Facilitate Knowledge Discovery**
    - Provide pre-defined question templates across policy categories.
    - Generate intelligent follow-up question suggestions.
    - Enable search within conversations.
6. **Support Organizational Learning**
    - Track analytics on policy queries and user engagement.
    - Identify knowledge gaps through user feedback.
    - Generate insights into most-accessed policy areas.
7. **Scalable Enterprise Knowledge Management**
    - Create a reusable framework applicable to other document types.
    - Support multi-user concurrent access.

4. Scope of Work

4.1 In Scope

The scope of this dissertation is limited to:

Backend Development

- API Design & Implementation: Developing a secure RESTful API using FastAPI, incorporating JWT authentication, RAG-based chat query processing, session management, and document serving.
- AI/ML Components: Integrating Google Gemini AI, implementing TF-IDF vector embedding generation for offline use, FAISS vector store, and advanced document retrieval with prompt engineering.
- Data Processing: Handling PDF parsing, intelligent document chunking, metadata extraction (categories, filenames, page numbers), and vector database initialization.
- Database Management: Designing and managing an SQLite database for user data, sessions, chat history, feedback, bookmarks, and notes with comprehensive schema design.

Frontend Development

- User Interface Components: Crafting a modern, responsive UI with a glass morphism design, animated chat interface, source document display with relevance scores, and dark/light theme support.
- User Features: Implementing secure login, comprehensive session management (create, rename, categorize, pin, archive), interactive message features (reactions, bookmarks, notes, copy), in-chat search, question templates, and follow-up suggestions.
- Admin Features: Developing an analytics dashboard to monitor usage statistics, user satisfaction metrics, and query volume trends.

4.2 Out of Scope

- Multi-language support: The current version is limited to English.
- Real-time collaboration features: No real-time co-editing or shared session functionalities.
- Policy document editing through the interface: The system is for retrieval, not content modification.
- Integration with external HR/IT systems: No direct links to other enterprise systems.
- Mobile native applications: Functionality is delivered via a web-based responsive design.
- Advanced analytics/BI dashboards beyond basic metrics: Focus on essential usage and satisfaction data.
- Multi-tenant architecture for serving multiple organizations: Designed for single-organization deployment.

5. Plan of Work

Phase	Start Date – End Date	Work to be Done
Phase 1: Foundation & Infrastructure	May 11 – May 31, 2026	Establish project foundation including repository setup, development environment, dependency management, and API structure using FastAPI. Implement core data processing capabilities for PDF parsing, text chunking, TF-IDF vector embedding generation, and FAISS index creation. Lay the groundwork for the React frontend, integrating Material-UI, routing, and basic authentication components. Achieve a fully functional document processing pipeline for 50+ PDFs and a secure authentication system. Draft Chapter 1 (Introduction)
Phase 2: RAG Implementation & Core Chat	June 1 – June 21, 2026	Integrate the Google Gemini API to build the Retrieval-Augmented Generation (RAG) service, focusing on document retrieval, context management, prompt engineering, and response generation with robust error handling. Develop essential chat API endpoints for querying, history, and session management. Construct a basic chat interface with message display, input functionality, source citations, and session listing. Conduct thorough unit and end-to-end testing to ensure high accuracy (>85%) and low response times (<3 seconds) with proper source attribution. Draft Chapter 2 (Literature Review)
Phase 3: Enhanced User Experience	June 22 – July 12, 2026	Implement advanced chat features such as message reactions, bookmarking, personal notes, copy-to-clipboard, in-chat search, and enhanced typing indicators.

		Improve session management with custom titles, categorization, pinning, archiving, and search functionality. Introduce user assistance features including question templates, follow-up suggestions, related policy recommendations, and document preview with page navigation. Refine the UI with a modern design, dark/light mode, smooth animations, and responsive optimization for an intuitive and accessible experience. Draft Chapter 3 (System Design)
Phase 4: Analytics and Admin Features	July 13 – July 31, 2026	Develop a comprehensive analytics dashboard to track total queries, users, sessions, and user satisfaction metrics through feedback analysis. Implement a robust feedback system with API endpoints and database schema for data collection and aggregation. Create personal bookmark management with search, filtering, and export capabilities. Introduce administrative controls for user management, system health monitoring, and database statistics, providing actionable insights into system usage and user behavior. Draft Chapter 4 (Implementation)
Phase 5: Testing and Deployment	Aug 1 - Aug 15, 2026	Prepare for deployment with environment configuration templates, database scripts, deployment instructions, security checklists, and backup procedures. Conduct rigorous unit / feature testing for all backend and frontend features. Draft Chapter 5 (Testing and Results)
Phase 6: Final Submission	August 15 – August 31, 2026	Deploy the project – Implement rollback mechanism and Docker caching. Draft Chapter 6 (Conclusion & Future Work). Compile all dissertation chapters. Create architecture diagrams. Prepare presentation Slides

6. Literature References

The following are the sources referred to from the preliminary literature review for this dissertation:

1. Lewis, P., Oguz, B., Yarats, D., Schick, T., Dwivedi, S., Xu, H., Golan, Y., Fan, A., Lu, M., Ritter, A., & Kiela, D. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. Advances in Neural Information Processing Systems, 33. https://arxiv.org/abs/2005.11401
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3. Liu, P., Yuan, W., Fu, J., Jiang, H., Hayashi, H., & Neubig, G. (2023). Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Pre-trained Language Models. *ACM Computing Surveys (CSUR), 55*(6), 1-38. https://arxiv.org/abs/2107.13586
4. Docker Inc. (2023). Docker Documentation. https://docs.docker.com/
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6. Google. Gemini Embeddings. https://ai.google.dev/gemini-api/docs/embeddings

7. Particulars of the Supervisor and Examiner

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8. Remarks of the Supervisor

The Maavu Policy Agent is a significant AI-powered RAG chatbot, transforming policy management by converting static documents into an interactive, intelligent system. This enhances accessibility, comprehension, and compliance.

The project commendably integrates NLP and full-stack development, delivering a practical, scalable, and privacy-conscious solution for enterprise information access.

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

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SEWIZC425T: DISSERTATION OUTLINE

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DISSERTATION / PROJECT / PROJECT WORK TITLE	:	**AI Powered Chatbot Agent for querying company policies using RAG and Google Gemini AI**

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